

MATH202-ENGINEERING MATHEMATICS

FALL2023

FINAL EXAM

TOTAL MARKS: 45 Points

DATE:14.12.2023

NAME: _____

TIME: 16:30 – 19:00

STUDENT ID: _____

SECTION: _____

INSTRUCTIONS:

1. This is a closed books/notes exam. Any notes, books, bags should be kept at the front of the classroom before the start of the test.
 2. Use of cellphones, laptops or any other communicating device is prohibited. Your cellphones should be switched off and submitted to the invigilator, or kept at the front of the class room before the start of the test.
 3. You are to attempt all questions on the answer sheet provided.
 4. Your solutions should be comprehensible. Explain what you are doing, and if you use any results that you have studied in this course or Calculus 1 and 2, mention them explicitly.
 5. Use of calculators is allowed for this exam.
 6. Any form of communication/sharing among your peers is not allowed during the exam.
 7. If you want to communicate with the invigilator, raise your hand.
 8. You are not allowed to leave your desk without permission of the invigilator during the exam.
 9. Failure to abide by the instructions above will lead to immediate cancellation of the exam.
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PROBLEM 1. (06 Points) Solve the following ODE

$$x \frac{dy}{dx} - 4y = x^6 e^x$$

for $0 < x < \infty$.

PROBLEM 2. (09 Points) Our goal is to solve the initial value problem:

$$y'' + y = 4x + 10 \sin x \tag{1}$$

with the initial conditions $y(\pi) = 0, y'(\pi) = 2$

- a. (3 Points) Find the general solution, y_c , of the associated homogeneous equation $y'' + y = 0$.
 - b. (2 Point) Suggest a guess function y_p for the non-homogeneous part $r(x) = 4x + 10 \sin x$. Justify your answer.
 - c. (3 Points) Solve for y_p using the Method of Undetermined Coefficients.
 - d. (0.5 Points) Write down the general solution of the second-order non-homogeneous equation (1).
 - e. (0.5 Points) Solve the initial value problem.
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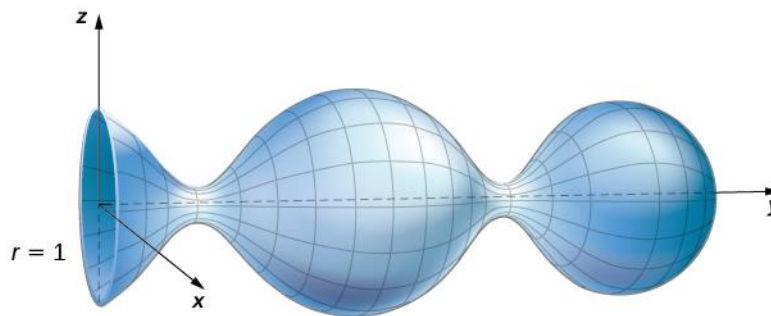
PROBLEM 3. (09 Points) Answer the questions below for the following initial value problem

$$y'' - 2xy' + 8y = 0, \quad y(0) = 3, y'(0) = 0$$

- (1 Point) Write down the guess for y as a power series centered at $x = 0$. Also determine y' and y'' as power series.
- (1 Point) Use the initial conditions to find the first two coefficients of the power series for y .
- (4 Points) Substitute your guess for y , y' , and y'' into the ODE and simplify.
- (2 Points) Find the recurrence relation for the coefficients
- (1 Point) Write down the general solution $y(x)$ as a power series.

PROBLEM 4. (05 Points) Let S be the smooth closed orientable boundary surface with enclosed volume V . Let a, b and c be arbitrary constants. Show that the total outward flux of the vector field $\vec{F}(x, y, z) = ax\hat{i} + by\hat{j} + cz\hat{k}$ through S , is equal to $(a + b + c)V$.

PROBLEM 5. (06 Points) Find $\iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} \, ds$ where $\vec{F}(x, y, z) = z\hat{i} + x\hat{j} + y\hat{k}$ and S is the surface as given in the figure.



PROBLEM 6. (02 Points each) For each of the statements below, indicate whether they are TRUE or FALSE. In each case provide a brief explanation or a counter example for your choice.

- The curl of a vector field is nonzero if and only if the vector field changes direction.
- The divergence points in the direction of steepest increase.
- If a particle moves along a curve C , the total work done W is independent of how quickly the particle moves.
- If y_1, y_2 and y_3 are three solutions of the equation $y''' + p(x)y'' + q(x)y' + r(x)y = 0$ then so is $c_1y_1 + c_2y_2 + c_3y_3$ where c_1, c_2 and c_3 are constants.
- The differential equation $(x - y)dx + xdy = 0$ is homogeneous.

END OF EXAM

In the elder days of Art,
Builders wrought with greatest care
Each minute and unseen part;
For the Gods are everywhere.

-Henry Wadsworth Longfellow

Important Formulae

- Assume sufficient differentiability of the functions and that the conditions for curves and surfaces are met, then:

Del Operator $\vec{\nabla}$	$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$
Gradient of a scalar function $\phi = \phi(x, y, z)$	$\vec{\nabla} \phi$
Divergence of a vector field $\vec{F}$	$\vec{\nabla} \cdot \vec{F}$
Curl of a vector field $\vec{F}$	$\vec{\nabla} \times \vec{F}$
Line integral of a vector field $\vec{F}$	$\int_C \vec{F} \cdot d\vec{r}$
Closed line integral	$\oint_C \vec{F} \cdot d\vec{r}$
Surface integral of vector field $\vec{F}$ over a surface S , where (u, v) are the parameters on the surface.	$\oiint_S \vec{F} \cdot \hat{n} dS = \iint_S \vec{F} \cdot \vec{N} du dv$

- For a point $P = (x, y, z)$ in Cartesian Coordinates, a representation in other coordinate systems can be

Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
$x = x$ $y = y$ $z = z$	$x = r \cos \theta$ $y = r \sin \theta$ $z = z$ where $r = \sqrt{x^2 + y^2} > 0$ $0 \leq \theta \leq 2\pi$	$x = \rho \sin \phi \cos \theta$ $y = \rho \sin \phi \sin \theta$ $z = \rho \cos \phi$ where $\rho = \sqrt{x^2 + y^2 + z^2} > 0$ $0 \leq \theta \leq 2\pi$ $0 \leq \phi \leq \pi$
$dV = dx dy dz$	$dV = r dr d\theta dz$	$dV = \rho^2 \sin \phi d\rho d\theta d\phi$

- Divergence theorem for a vector field $\vec{F}$,

$$\iiint_D \vec{\nabla} \cdot \vec{F} dV = \oiint_S \vec{F} \cdot \hat{n} ds$$

Here, D is the region enclosed by the closed positively oriented surface S and V is its volume.

- Stoke's Theorem for a vector field $\vec{F}$,

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} \, ds = \oint_C \vec{F} \cdot d\vec{r}$$

Here the closed positively oriented curve C bounds the surface S in 3-dimensional space.

- A first order linear ODE $y' + p(x)y = r(x)$ has solution

$$y = e^{-\int p(x)dx} \left(\int e^{\int p(x)dx} r(x) + C \right)$$

- A first order ODE $M(x,y)dx + N(x,y)dy = 0$ is exact iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

- A first order non-exact ODE, $P(x,y) dx + Q(x,y) dy = 0$, can be made exact by multiplying it with an integrating function $F(x)$:

$$F(x) = e^{\int R(x) dx}, \text{ where } R(x) = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right).$$

if such a function exists.

Or, multiplying it with an integrating function $G(y)$:

$$G(y) = e^{\int R^*(y) dy}, \text{ where } R^*(y) = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right).$$

if such a function exists.

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- Function guesses for Method of Undetermined Coefficient,

$S(x)$	Guess for $y_p(x)$
$k = \text{Constant}$	K (Some other constant)
ke^{rx}	Ke^{rx}
kx^n, ($n = \text{positive integer}$)	$K_n x^n + K_{n-1} x^{n-1} + \dots + K_1 x + K_0$
$k \sin \omega x$	$K \cos \omega x + M \sin \omega x$
$k \cos \omega x$	
$ke^{\alpha x} \cos \omega x$	$e^{\alpha x} (K \cos \omega x + M \sin \omega x)$
$ke^{\alpha x} \sin \omega x$	

Table 1: Short list of guesses of particular solutions for some forms of non-homogeneous part.

- The power series of $y(x)$ about $x_0 = 0$,

$$y(x) = a_0 + a_1(x) + a_2(x)^2 + a_3(x)^3 + \dots = \sum_{n=0}^{\infty} a_n x^n$$
